Cognitive Robotics Lab – 7

23BRS1177

GitHub: https://github.com/Prayagraj27/Cognitive-Robotics-Lab

Obstacle Avoidance using SLAM

Using SLAM by integrating LIDAR data with wheel encoder data to map objects and simultaneously localize itself.

Code

```
import math
import time
import sys
import signal
import rclpy
from rclpy.node import Node
from nav_msgs.msg import OccupancyGrid
from sensor_msgs.msg import LaserScan
from geometry_msgs.msg import Twist
import tf2_ros
from tf2_ros import TransformException

class MapAvoidance(Node):
    def __init__(self):
        super().__init__('map_avoidance')
        self.declare_parameter('enable_motion', False)
        self.declare_parameter('map_topic', '/map')
        self.declare_parameter('scan_topic', '/scan')
        self.declare_parameter('cmd_vel_topic', '/cmd_vel_nav')
        self.declare_parameter('map_frame', 'map')
        self.declare_parameter('base_frame', 'base_link')
        self.declare_parameter('emergency_stop_distance', 1/5)
        self.declare_parameter('safe_distance', 7/20)
        self.declare_parameter('map_check_distance', 1/2)
        self.declare_parameter('forward_width', 3/10)
        self.declare_parameter('occupied_threshold', 50)
        self.declare_parameter('unknown_is_blocked', False)
        self.declare_parameter('forward_speed', 3/25)
        self.declare_parameter('control_period', 1/10)
```

```
        self.enable_motion = bool(self.get_parameter('enable_motion').value)
        self.map_topic = self.get_parameter('map_topic').value
        self.scan_topic = self.get_parameter('scan_topic').value
        self.cmd_vel_topic = self.get_parameter('cmd_vel_topic').value
        self.map_frame = self.get_parameter('map_frame').value
        self.base_frame = self.get_parameter('base_frame').value
        self.emergency_stop_distance =
float(self.get_parameter('emergency_stop_distance').value)
        self.safe_distance = float(self.get_parameter('safe_distance').value)
        self.map_check_distance =
float(self.get_parameter('map_check_distance').value)
        self.forward_width = float(self.get_parameter('forward_width').value)
        self.occupied_threshold =
int(self.get_parameter('occupied_threshold').value)
        self.unknown_is_blocked =
bool(self.get_parameter('unknown_is_blocked').value)
        self.forward_speed = float(self.get_parameter('forward_speed').value)
        self.control_period = float(self.get_parameter('control_period').value)

        self.map_msg = None
        self.scan_msg = None
        self.last_map_time = 0.0
        self.last_scan_time = 0.0
        self.last_decision = ""

        self.map_sub = self.create_subscription(OccupancyGrid, self.map_topic,
self.map_callback, 10)
        self.scan_sub = self.create_subscription(LaserScan, self.scan_topic,
self.scan_callback, 10)
        self.cmd_pub = self.create_publisher(Twist, self.cmd_vel_topic, 10)
        self.tf_buffer = tf2_ros.Buffer()
        self.tf_listener = tf2_ros.TransformListener(self.tf_buffer, self)
        self.timer = self.create_timer(self.control_period, self.control_loop)

        signal.signal(signal.SIGINT, self.signal_handler)

        self.get_logger().info('================================================
===')
        self.get_logger().info('LYRA MAP AVOIDANCE STARTED')
        self.get_logger().info(f'Motion enabled: {self.enable_motion}')
        self.get_logger().info('================================================
===')

    def signal_handler(self, sig, frame):
        self.get_logger().info('CRITICAL INTERCEPT: Halting wheels before exit')
```

```
        for _ in range(20):
            self.stop_robot()
            time.sleep(1/50)
        self.destroy_node()
        rclpy.try_shutdown()
        sys.exit(0)

    def map_callback(self, msg):
        self.map_msg = msg
        self.last_map_time = time.monotonic()

    def scan_callback(self, msg):
        self.scan_msg = msg
        self.last_scan_time = time.monotonic()

    def stop_robot(self):
        msg = Twist()
        msg.linear.x = 0.0
        msg.linear.y = 0.0
        msg.linear.z = 0.0
        msg.angular.x = 0.0
        msg.angular.y = 0.0
        msg.angular.z = 0.0
        self.cmd_pub.publish(msg)

    def get_robot_pose(self):
        try:
            transform = self.tf_buffer.lookup_transform(self.map_frame,
self.base_frame, rclpy.time.Time())
            x = transform.transform.translation.x
            y = transform.transform.translation.y
            qx = transform.transform.rotation.x
            qy = transform.transform.rotation.y
            qz = transform.transform.rotation.z
            qw = transform.transform.rotation.w
            siny_cosp = 2.0 * (qw * qz + qx * qy)
            cosy_cosp = 1.0 - 2.0 * (qy * qy + qz * qz)
            yaw = math.atan2(siny_cosp, cosy_cosp)
            return x, y, yaw
        except TransformException as ex:
            self.get_logger().warn(f'Waiting for TF {self.map_frame} ->
{self.base_frame}', throttle_duration_sec=3.0)
            return None

    def world_to_map(self, x, y):
```

```
        if self.map_msg is None:
            return None
        info = self.map_msg.info
        resolution = info.resolution
        origin_x = info.origin.position.x
        origin_y = info.origin.position.y
        cell_x = int(math.floor((x - origin_x) / resolution))
        cell_y = int(math.floor((y - origin_y) / resolution))
        if cell_x < 0:
            return None
        if cell_y < 0:
            return None
        if cell_x >= info.width:
            return None
        if cell_y >= info.height:
            return None
        return cell_x, cell_y

    def get_map_value(self, cell_x, cell_y):
        if self.map_msg is None:
            return None
        info = self.map_msg.info
        if cell_x < 0 or cell_x >= info.width:
            return None
        if cell_y < 0 or cell_y >= info.height:
            return None
        index = cell_y * info.width + cell_x
        if index < 0:
            return None
        if index >= len(self.map_msg.data):
            return None
        return self.map_msg.data[index]

    def is_cell_safe(self, value):
        if value is None:
            return False
        if value < 0:
            if self.unknown_is_blocked:
                return False
            return True
        if value >= self.occupied_threshold:
            return False
        return True

    def check_map_ahead(self, robot_x, robot_y, yaw):
```

```
        if self.map_msg is None:
            return False, 0.0, 0
        resolution = self.map_msg.info.resolution
        distance_steps = max(1, int(self.map_check_distance / resolution))
        width_steps = max(0, int((self.forward_width / 2.0) / resolution))
        checked = 0
        blocked = 0
        minimum_free_distance = self.map_check_distance
        for distance_index in range(1, distance_steps + 1):
            distance = distance_index * resolution
            center_x = robot_x + math.cos(yaw) * distance
            center_y = robot_y + math.sin(yaw) * distance
            perp_x = -math.sin(yaw)
            perp_y = math.cos(yaw)
            row_blocked = False
            for width_index in range(-width_steps, width_steps + 1):
                lateral = width_index * resolution
                sample_x = center_x + perp_x * lateral
                sample_y = center_y + perp_y * lateral
                cell = self.world_to_map(sample_x, sample_y)
                if cell is None:
                    row_blocked = True
                    checked += 1
                    continue
                cell_x, cell_y = cell
                value = self.get_map_value(cell_x, cell_y)
                checked += 1
                if not self.is_cell_safe(value):
                    row_blocked = True
            if row_blocked:
                minimum_free_distance = distance
                blocked += 1
                return False, minimum_free_distance, checked
        return True, minimum_free_distance, checked

    def get_lidar_front_distance(self):
        if self.scan_msg is None:
            return None
        scan = self.scan_msg
        minimum = float('inf')
        half_width = math.radians(20.0)
        for i, distance in enumerate(scan.ranges):
            if not math.isfinite(distance):
                continue
            if distance < scan.range_min:
```

```
                continue
            if distance > scan.range_max:
                continue
            angle = scan.angle_min + i * scan.angle_increment
            angle = math.atan2(math.sin(angle), math.cos(angle))
            difference = abs(abs(angle) - math.pi)
            if difference <= half_width:
                if distance < minimum:
                    minimum = distance
        if math.isfinite(minimum):
            return minimum
        return None

    def control_loop(self):
        if not self.enable_motion:
            self.stop_robot()
            if self.last_decision != 'DISABLED':
                self.get_logger().info('Motion disabled.')
                self.last_decision = 'DISABLED'
            return

        if self.map_msg is None:
            self.stop_robot()
            if self.last_decision != 'WAIT_MAP':
                self.get_logger().info('Waiting for /map...')
                self.last_decision = 'WAIT_MAP'
            return

        if self.scan_msg is None:
            self.stop_robot()
            if self.last_decision != 'WAIT_SCAN':
                self.get_logger().info('Waiting for /scan...')
                self.last_decision = 'WAIT_SCAN'
            return

        pose = self.get_robot_pose()
        if pose is None:
            self.stop_robot()
            return

        robot_x, robot_y, yaw = pose

        lidar_front = self.get_lidar_front_distance()
        if lidar_front is None:
            self.stop_robot()
```

```
            return

        if lidar_front <= self.emergency_stop_distance:
            self.stop_robot()
            self.get_logger().error(f'EMERGENCY STOP: LiDAR obstacle at
{lidar_front} m')
            self.last_decision = 'EMERGENCY_STOP'
            return

        map_safe, map_distance, checked = self.check_map_ahead(robot_x, robot_y,
yaw)

        if not map_safe:
            cmd = Twist()
            cmd.linear.x = 0.0
            cmd.angular.z = float(7/20)
            self.cmd_pub.publish(cmd)
            self.get_logger().warn(f'MAP BLOCKED: map={map_distance}m.
Pivoting.')
            self.last_decision = 'MAP_STOP'
            return

        if lidar_front <= self.safe_distance:
            cmd = Twist()
            cmd.linear.x = 0.0
            cmd.angular.z = float(7/20)
            self.cmd_pub.publish(cmd)
            self.get_logger().warn(f'LiDAR STOP: obstacle at {lidar_front}m.
Pivoting.')
            self.last_decision = 'LIDAR_STOP'
            return

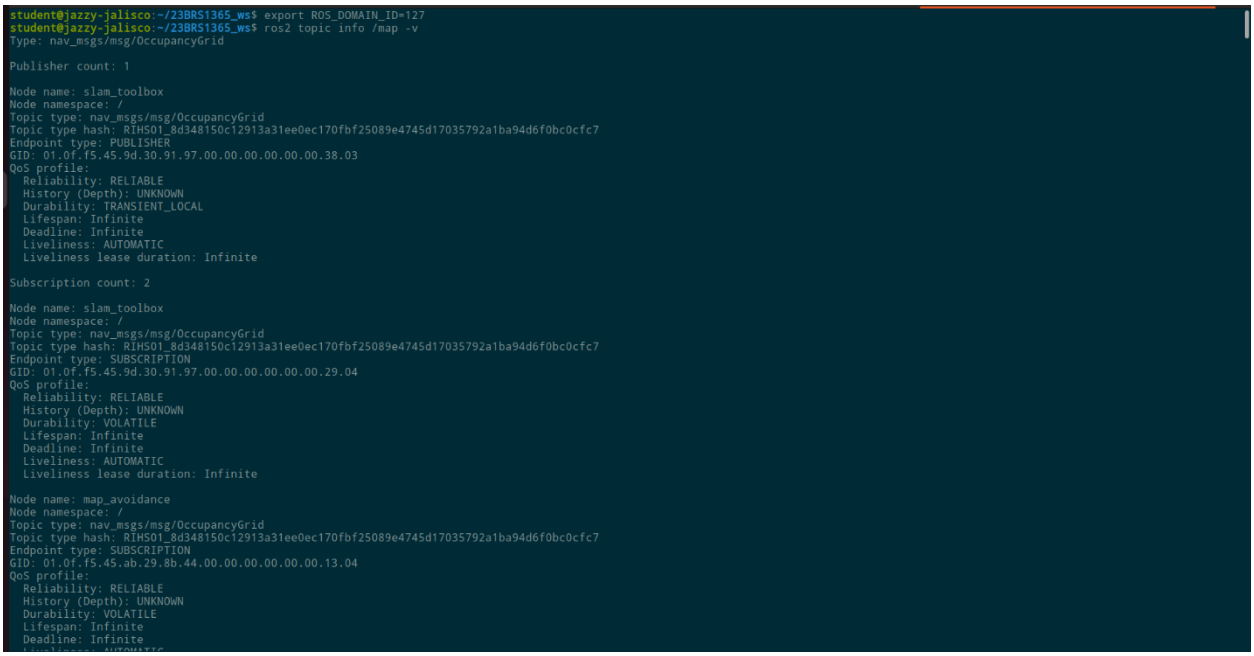
        cmd = Twist()
        cmd.linear.x = self.forward_speed
        cmd.angular.z = 0.0
        self.cmd_pub.publish(cmd)
        self.get_logger().info(f'DECISION: FORWARD | Map={map_distance}m |
LiDAR={lidar_front}m')
        self.last_decision = 'FORWARD'

def main(args=None):
    rclpy.init(args=args)
    node = MapAvoidance()
    try:
        rclpy.spin(node)
```

```
    except KeyboardInterrupt:
        pass
    finally:
        rclpy.try_shutdown()

if __name__ == '__main__':
    main()
```

Output





```
student@jazzy-jalisco:~$ ros2 run tf2_ros tf2_echo odom base_link
[INFO] [1791445465.570441070] [tf2_echo]: Waiting for transform odom ->  base_link: Invalid frame ID "odom" passed to canTransform argument target_frame - frame does not exist
[INFO] [1791445467.561280992] [tf2_echo]: Waiting for transform odom ->  base_link: Invalid frame ID "odom" passed to canTransform argument target_frame - frame does not exist
At time 1791445468.551051465
- Translation: [0.810, -0.201, 0.163]
- Rotation: in Quaternion (xyzw) [0.000, 0.000, -0.426, 0.905]
- Rotation: in RPY (radian) [0.000, 0.000, -0.880]
- Rotation: in RPY (degree) [0.000, 0.000, -50.430]
- Matrix:
  0.637  0.771  0.000  0.810
 -0.771  0.637 -0.000 -0.201
 -0.000  0.000  1.000  0.163
  0.000  0.000  0.000  1.000
At time 1791445469.548174287
- Translation: [0.767, -0.193, 0.163]
- Rotation: in Quaternion (xyzw) [0.000, 0.000, -0.424, 0.905]
- Rotation: in RPY (radian) [0.000, 0.000, -0.877]
- Rotation: in RPY (degree) [0.000, 0.000, -50.222]
- Matrix:
  0.640  0.769  0.000  0.767
 -0.769  0.640 -0.000 -0.193
 -0.000  0.000  1.000  0.163
  0.000  0.000  0.000  1.000
At time 1791445470.544846214
- Translation: [0.810, -0.201, 0.163]
- Rotation: in Quaternion (xyzw) [0.000, 0.000, -0.428, 0.904]
- Rotation: in RPY (radian) [0.000, 0.000, -0.885]
- Rotation: in RPY (degree) [0.000, 0.000, -50.714]
- Matrix:
  0.633  0.774  0.000  0.810
 -0.774  0.633 -0.000 -0.201
 -0.000  0.000  1.000  0.163
  0.000  0.000  0.000  1.000
At time 1791445471.543084803
- Translation: [0.810, -0.201, 0.163]
- Rotation: in Quaternion (xyzw) [0.000, 0.000, -0.428, 0.904]
- Rotation: in RPY (radian) [0.000, 0.000, -0.884]
- Rotation: in RPY (degree) [0.000, 0.000, -50.661]
- Matrix:
  0.634  0.773  0.000  0.810
 -0.773  0.634 -0.000 -0.201
 -0.000  0.000  1.000  0.163
  0.000  0.000  0.000  1.000
At time 1791445472.545000029
- Translation: [0.809, -0.201, 0.163]
- Rotation: in Quaternion (xyzw) [0.000, 0.000, -0.423, 0.906]
- Rotation: in RPY (radian) [0.000, 0.000, -0.874]
- Rotation: in RPY (degree) [0.000, 0.000, -50.052]
- Matrix:
```

```
^Cstudent@jazzy-jalisco:~$ ros2 topic hz /wheel/odom
WARNING: topic [/wheel/odom] does not appear to be published yet
average rate: 10.016
        min: 0.079s max: 0.109s std dev: 0.00849s window: 11
average rate: 10.406
        min: 0.003s max: 0.115s std dev: 0.02194s window: 23
average rate: 13.682
        min: 0.000s max: 0.115s std dev: 0.04411s window: 45
average rate: 15.137
        min: 0.000s max: 0.115s std dev: 0.04675s window: 65
average rate: 16.230
        min: 0.000s max: 0.115s std dev: 0.04801s window: 86
average rate: 16.753
        min: 0.000s max: 0.176s std dev: 0.04941s window: 107
^Cstudent@jazzy-jalisco:~ros2 topic hz /odometry/filtered
WARNING: topic [/odometry/filtered] does not appear to be published yet
average rate: 37.330
        min: 0.000s max: 0.124s std dev: 0.03197s window: 41
average rate: 38.561
        min: 0.000s max: 0.124s std dev: 0.02846s window: 81
average rate: 39.022
        min: 0.000s max: 0.124s std dev: 0.02682s window: 123
average rate: 39.496
        min: 0.000s max: 0.124s std dev: 0.02632s window: 164
^Cstudent@jazzy-jalisco:~ros2 topic info /imu/data -vv
Type: sensor_msgs/msg/Imu

Publisher count: 1

Node name: imu_filter
Node namespace: /
Topic type: sensor_msgs/msg/Imu
Topic type hash: RIHS01_7d9a00ff131080897a5ec7e26e315954b8eae3353c3f995c55faf71574000b5b
Endpoint type: PUBLISHER
GID: 01.0f.8a.77.2b.18.69.2a.00.00.00.00.00.00.22.03
QoS profile:
  Reliability: RELIABLE
  History (Depth): UNKNOWN
  Durability: VOLATILE
  Lifespan: Infinite
  Deadline: Infinite
  Liveliness: AUTOMATIC
  Liveliness lease duration: Infinite

Subscription count: 2

Node name: ekf_filter_node
Node namespace: /
Topic type: sensor_msgs/msg/Imu
Topic type hash: RIHS01_7d9a00ff131080897a5ec7e26e315954b8eae3353c3f995c55faf71574000b5b
```

```
[WARN] [1791445931.421573080] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.97 m
[WARN] [1791445931.521582170] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.97 m
[WARN] [1791445931.621591523] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.10 m | LiDAR=0.97 m
[WARN] [1791445931.721533452] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.15 m | LiDAR=0.97 m
[WARN] [1791445931.821512955] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.96 m
[WARN] [1791445931.921496740] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.96 m
[WARN] [1791445932.021611844] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.94 m
[WARN] [1791445932.121538290] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.90 m
[WARN] [1791445932.221521026] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.84 m
[WARN] [1791445932.321521328] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.10 m | LiDAR=0.75 m
[WARN] [1791445932.421686362] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.15 m | LiDAR=0.71 m
[WARN] [1791445932.521501180] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.69 m
[WARN] [1791445932.621485693] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.68 m
[WARN] [1791445932.721533302] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.67 m
[WARN] [1791445932.821696684] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.67 m
[WARN] [1791445932.921617964] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.15 m | LiDAR=0.67 m
[WARN] [1791445933.021501535] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.67 m
[WARN] [1791445933.121546558] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.67 m
[WARN] [1791445933.221555638] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.65 m
[WARN] [1791445933.321586505] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.55 m
[WARN] [1791445933.421564525] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.46 m
[WARN] [1791445933.521541381] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.43 m
[WARN] [1791445933.621574921] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.10 m | LiDAR=0.42 m
[WARN] [1791445933.721503519] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.41 m
[WARN] [1791445933.821529456] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.41 m
[WARN] [1791445933.921544376] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.40 m
[WARN] [1791445934.021524445] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.40 m
[WARN] [1791445934.121550030] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.40 m
[WARN] [1791445934.221544217] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.41 m
[WARN] [1791445934.321606601] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.40 m
[WARN] [1791445934.421558630] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.40 m
[WARN] [1791445934.521497333] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.40 m
[WARN] [1791445934.621609089] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.40 m
[WARN] [1791445934.722265887] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.38 m
[WARN] [1791445934.822137846] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.35 m
[WARN] [1791445934.921738313] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.34 m
[WARN] [1791445935.021571283] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.34 m
[WARN] [1791445935.121589314] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.34 m
[WARN] [1791445935.222036472] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.34 m
[WARN] [1791445935.321601035] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.38 m
[WARN] [1791445935.421459337] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.38 m
[WARN] [1791445935.521007088] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.37 m
[WARN] [1791445935.621554399] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.37 m
[WARN] [1791445935.721550506] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.37 m
[WARN] [1791445935.821828227] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.37 m
[WARN] [1791445935.921548148] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.38 m
[WARN] [1791445936.021596291] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.37 m
[WARN] [1791445936.121551501] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.36 m
[WARN] [1791445936.221569784] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.36 m
[WARN] [1791445936.321782907] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=0.36 m
```

```
[WARN] [1791445843.121740382] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.40 m
[WARN] [1791445843.221705834] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.41 m
[WARN] [1791445843.321796294] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.40 m
[WARN] [1791445843.421782992] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.40 m
[WARN] [1791445843.521727349] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.40 m
[WARN] [1791445843.621800115] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=2.29 m
[WARN] [1791445843.721689228] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.40 m
[ERROR] [1791445843.821427643] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.06 m
[ERROR] [1791445843.921587908] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.05 m
[ERROR] [1791445844.021682385] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.05 m
[ERROR] [1791445844.121706515] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.06 m
[ERROR] [1791445844.221639185] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.06 m
[ERROR] [1791445844.321481351] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.06 m
[ERROR] [1791445844.421533081] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.06 m
[ERROR] [1791445844.521671440] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.06 m
[ERROR] [1791445844.621449161] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.06 m
[ERROR] [1791445844.721550652] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.05 m
[ERROR] [1791445844.821435046] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.06 m
[ERROR] [1791445844.921639454] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.06 m
[ERROR] [1791445845.021545380] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.05 m
[ERROR] [1791445845.121709126] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.06 m
[ERROR] [1791445845.231067534] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.06 m
[ERROR] [1791445845.321526760] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.05 m
[ERROR] [1791445845.422043635] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.06 m
[ERROR] [1791445845.521636433] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.05 m
[ERROR] [1791445845.621559920] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.05 m
[ERROR] [1791445845.721604744] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.05 m
[ERROR] [1791445845.821705490] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.05 m
[ERROR] [1791445845.920686156] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.05 m
[ERROR] [1791445846.021619563] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.05 m
[ERROR] [1791445846.122015780] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.05 m
[ERROR] [1791445846.221636288] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.05 m
[ERROR] [1791445846.386827633] [map_avoidance]: EMERGENCY STOP: LiDAR obstacle at 0.05 m
[WARN] [1791445846.421791760] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.40 m
[WARN] [1791445846.521628729] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=2.29 m
[WARN] [1791445846.621653349] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.40 m
[WARN] [1791445846.721685937] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.40 m
[WARN] [1791445846.821627213] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=2.29 m
[WARN] [1791445846.921742858] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.20 m | LiDAR=1.39 m
[WARN] [1791445847.098325958] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=2.29 m
[WARN] [1791445847.121719102] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.40 m
[WARN] [1791445847.221751586] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.40 m
[WARN] [1791445847.321712568] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=2.29 m
[WARN] [1791445847.421752572] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.40 m
[WARN] [1791445847.520884970] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.40 m
[WARN] [1791445847.621760302] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=2.29 m
[WARN] [1791445847.721662234] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.40 m
[WARN] [1791445847.821681710] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=1.40 m
[WARN] [1791445847.921639190] [map_avoidance]: MAP BLOCKED: forward path blocked/unknown at approximately 0.05 m | LiDAR=2.29 m
```

```
student@jazzy-jalisco:~/23BRS1365_ws$ cd /src/SLAM
bash: cd: /src/SLAM: No such file or directory
student@jazzy-jalisco:~/23BRS1365_ws$ cd src
student@jazzy-jalisco:~/23BRS1365_ws/src$ ls
build  install  log  SLAM
student@jazzy-jalisco:~/23BRS1365_ws/src$ cd SLAM
student@jazzy-jalisco:~/23BRS1365_ws/src/SLAM$ ls
CMakeLists.txt  include  map_avoidance.py  package.xml  src
student@jazzy-jalisco:~/23BRS1365_ws/src/SLAM$ ros2 topic echo /map
header:
  stamp:
    sec: 1791445625
    nanosec: 381285719
  frame_id: map
info:
  map_load_time:
    sec: 0
    nanosec: 0
  resolution: 0.05000000074505806
  width: 153
  height: 142
  origin:
    position:
      x: -3.6036624226827763
      y: -2.822906718833629
      z: 0.0
    orientation:
      x: 0.0
      y: 0.0
      z: 0.0
      w: 1.0
data:
- -1
- -1
- -1
- -1
- -1
- -1
- -1
- -1
- -1
- -1
- -1
- -1
- -1
- -1
```

```
student@jazzy-jalisco:~/23BRS1365_ws$ export ROS_DOMAIN_ID=127
student@jazzy-jalisco:~/23BRS1365_ws$ ros2 topic info /map -v
Type: nav_msgs/msg/OccupancyGrid

Publisher count: 1

Node name: slam_toolbox
Node namespace: /
Topic type: nav_msgs/msg/OccupancyGrid
Topic type hash: RIHS01_8d348150c12913a31ee0ec170fbf25089e4745d17035792a1ba94d6f0bc0cfc7
Endpoint type: PUBLISHER
GID: 01.0f.f5.45.9d.30.91.97.00.00.00.00.00.00.38.03
QoS profile:
  Reliability: RELIABLE
  History (Depth): UNKNOWN
  Durability: TRANSIENT_LOCAL
  Lifespan: Infinite
  Deadline: Infinite
  Liveliness: AUTOMATIC
  Liveliness lease duration: Infinite

Subscription count: 2

Node name: slam_toolbox
Node namespace: /
Topic type: nav_msgs/msg/OccupancyGrid
Topic type hash: RIHS01_8d348150c12913a31ee0ec170fbf25089e4745d17035792a1ba94d6f0bc0cfc7
Endpoint type: SUBSCRIPTION
GID: 01.0f.f5.45.9d.30.91.97.00.00.00.00.00.00.29.04
QoS profile:
  Reliability: RELIABLE
  History (Depth): UNKNOWN
  Durability: VOLATILE
  Lifespan: Infinite
  Deadline: Infinite
  Liveliness: AUTOMATIC
  Liveliness lease duration: Infinite

Node name: map_avoidance
Node namespace: /
Topic type: nav_msgs/msg/OccupancyGrid
Topic type hash: RIHS01_8d348150c12913a31ee0ec170fbf25089e4745d17035792a1ba94d6f0bc0cfc7
Endpoint type: SUBSCRIPTION
GID: 01.0f.f5.45.ab.29.8b.44.00.00.00.00.00.00.13.04
QoS profile:
  Reliability: RELIABLE
  History (Depth): UNKNOWN
  Durability: VOLATILE
  Lifespan: Infinite
  Deadline: Infinite
```